

Day 26: Performance Optimization & VertiPaq Engine

[DEF] Definition to Remember

Performance Optimization in Power BI involves reducing semantic model memory footprint and tuning DAX queries to ensure visual rendering completes in under 1-2 seconds.

1. Using the Performance Analyzer

- * Built-in diagnostic tool (View -> Performance Analyzer) recording 3 visual cost components:

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2. Column Cardinality & VertiPaq Compression

- * VertiPaq stores data column-by-column using Dictionary, Run-Length (RLE), and Bit-Packed compression.

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3. DAX Optimization Best Practices

- * Avoid iterating full tables with FILTER(AllTable, ...) when simple column filters suffice.

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[MUST-WRITE] Must-Master Skills & Interview Points

1. How to diagnose visual rendering lag with Performance Analyzer.
2. What column cardinality is and why reducing it drastically shrinks .pbix file size.
3. Using DAX Studio and Tabular Editor for advanced query plan and VertiPaq metrics analysis.

[RECALL] Quick Recall

Performance Analyzer + Reduce Column Cardinality + Split DateTimes + Cache VARs in DAX